

Saddle-Dominated Capacity of the LSE Dense Associative Memory

Abstract

[Placeholder.] The log-sum-exp Dense Associative Memory [3] stores M unit vectors ξ^μ on the N -sphere with kernel sharpness β_{net} . Ramsauer et al. proved $\alpha_c = 1/2$ at $T = 0$ by modelling the inter-pattern overlaps as Gaussian. Replacing the Gaussian by the exact spherical density opens an interior saddle in the partition sum at $\varphi^*(\beta_{\text{net}})$, with value $g_{\text{max}}(\beta_{\text{net}})$. The boundary becomes $\alpha_c(\beta_{\text{net}}, T) = \beta_{\text{net}}[1 - f_{\text{ret}}(T)] - g_{\text{max}}(\beta_{\text{net}})$. At $\beta_{\text{net}} = 1$ — the value we use in the Monte Carlo — the saddle is the golden ratio $\varphi^* = (\sqrt{5} - 1)/2$ and $g_{\text{max}} \approx 0.3774$, giving $\alpha_c(0) \approx 0.6226$, not 0.5. Ramsauer’s $1/2$ is β_{net} -independent (an extreme-value cancellation), the exact-density boundary is not. We confirm the corrected value with three Monte Carlo variants (direct, truncated, hybrid) at N up to 100.

1 Introduction

A Dense Associative Memory [1] stores M patterns $\xi^\mu \in \mathbb{R}^N$, $\mu = 1, \dots, M$, as minima of an energy function $H(\mathbf{x})$. Retrieval of ξ^1 from a noisy cue is the Metropolis dynamics of $\mathbf{x}$ under H at external temperature T . Both the patterns and the chain live on the N -sphere of radius $\sqrt{N}$.

The log-sum-exp (LSE) variant of Ramsauer et al. uses

$$H(\mathbf{x}) = -\beta_{\text{net}}^{-1} \ln \sum_{\mu=1}^M \exp(\beta_{\text{net}} \mathbf{x} \cdot \xi^\mu), \quad (1)$$

with β_{net} the kernel sharpness (inverse variance, unrelated to the external temperature T), and stores $M = e^{\alpha N}$ patterns at load $\alpha = \ln M/N$. The question is: what is the largest α at which the chain initialised at ξ^1 stays close to ξ^1 ?

Ramsauer et al. answered $\alpha_c = 1/2$ at $T = 0$. The proof modelled the inter-pattern overlap $\varphi_{1\mu} = \xi^1 \cdot \xi^\mu / N$ as a Gaussian with variance $1/N$. The Gaussian is the leading term of the exact spherical density

$$f(\varphi) \propto (1 - \varphi^2)^{(N-3)/2}, \quad \varphi \in [-1, 1], \quad (2)$$

near $\varphi = 0$, but the two densities differ near $\varphi = 1$. That tail matters: it sets the typical maximum of M samples and the saddle of the partition sum.

This paper does three things. First, we re-derive the boundary using the exact density (2). An interior saddle of the integrand appears at a point $\varphi^*(\beta_{\text{net}})$ that depends on the kernel sharpness, with saddle value $g_{\text{max}}(\beta_{\text{net}})$. At $\beta_{\text{net}} = 1$ — the value used in the Monte Carlo of Sections 3 and 4 — the saddle is the golden ratio $\varphi^* = (\sqrt{5} - 1)/2$, $g_{\text{max}} \approx 0.3774$, and the boundary shifts to $\alpha_c(0) = 1 - g_{\text{max}} \approx 0.6226$. Second, we describe three Monte Carlo schemes (direct, truncated, hybrid) and explain why a naive “smart” variant that refreshes its pattern cache fails near α_c . Third, we confirm the corrected boundary numerically at $N = 100$ by an escape-time measurement and by a ϕ residual heatmap.

2 Theory

Free energy split. For a chain in the basin of ξ^1 , the LSE sum is dominated by the $\mu = 1$ term, $H(\mathbf{x})/N \approx -\beta_{\text{net}}^{-1} \ln e^{\beta_{\text{net}} N \varphi_1} = -\varphi_1$ with $\varphi_1 = \xi^1 \cdot \mathbf{x}/N$ — the β_{net} factor cancels for a

single-pattern dominant term, so H/N at retrieval is β_{net} -independent. Define the energy per spin relative to the retrieval baseline, $u(\varphi_1) = 1 - \varphi_1$, and the $(N-1)$ -sphere entropy per spin $s(\varphi_1) = \frac{1}{2} \ln(1 - \varphi_1^2)$. The retrieval free energy is the minimum of $f(\varphi_1, T) = u - Ts$ (also β_{net} -independent for the same reason):

$$f_{\text{ret}}(T) = (1 - \phi_{\text{eq}}) - \frac{T}{2} \ln(1 - \phi_{\text{eq}}^2), \quad \phi_{\text{eq}}(T) = \frac{1}{2}(-T + \sqrt{T^2 + 4}). \quad (3)$$

For a chain uncorrelated with ξ^1 ($\varphi_1 \approx 0$) the sum in (1) is dominated by the $M-1$ spurious terms $S = \sum_{\mu \neq 1} \exp(\beta_{\text{net}} N \varphi_{1\mu})$. Replacing the sum by an integral with density (2) and using $M = e^{\alpha N}$:

$$S \approx e^{\alpha N} \int_{-1}^1 \exp\left[N\left(\frac{1}{2} \ln(1 - \varphi^2) + \beta_{\text{net}} \varphi\right)\right] d\varphi \equiv e^{\alpha N} \int e^{Ng(\varphi; \beta_{\text{net}})} d\varphi. \quad (4)$$

We evaluate (4) by the saddle-point method in the large- N limit (Laplace's method): the integrand concentrates at the maximiser $\varphi^*(\beta_{\text{net}})$ of the exponent $g(\varphi; \beta_{\text{net}}) = \frac{1}{2} \ln(1 - \varphi^2) + \beta_{\text{net}} \varphi$, and

$$\int e^{Ng(\varphi; \beta_{\text{net}})} d\varphi \sim \sqrt{\frac{2\pi}{N|g''(\varphi^*)|}} e^{Ng(\varphi^*; \beta_{\text{net}})} \implies \frac{1}{N} \ln S \rightarrow \alpha + g_{\text{max}}(\beta_{\text{net}}) \quad (N \rightarrow \infty), \quad (5)$$

with subleading $\log N/N$ corrections. The saddle condition $g'(\varphi^*) = 0$ gives $\beta_{\text{net}}(1 - \varphi^{*2}) = \varphi^*$, with positive root

$$\varphi^*(\beta_{\text{net}}) = \frac{-1 + \sqrt{1 + 4\beta_{\text{net}}^2}}{2\beta_{\text{net}}}. \quad (6)$$

The saddle value is

$$g_{\text{max}}(\beta_{\text{net}}) \equiv g(\varphi^*; \beta_{\text{net}}) = \frac{1}{2} \ln(1 - \varphi^{*2}) + \beta_{\text{net}} \varphi^*. \quad (7)$$

The noise-floor energy per spin is $u_{\text{noise}} = 1 - [\alpha + g_{\text{max}}(\beta_{\text{net}})]/\beta_{\text{net}}$, and the boundary at the crossover $u_{\text{noise}} = f_{\text{ret}}(T)$ is

$$\boxed{\alpha_c(\beta_{\text{net}}, T) = \beta_{\text{net}} [1 - f_{\text{ret}}(T)] - g_{\text{max}}(\beta_{\text{net}}).} \quad (8)$$

Gaussian and support-edge variants. Two well-known prior formulas come from substituting different objects into (4) and using its *support edge* instead of the interior saddle.

Gaussian density. Replace $f(\varphi)$ by the moment-matched Gaussian $\sqrt{N/2\pi} \exp(-N\varphi^2/2)$. The integrand exponent becomes $g_G(\varphi; \beta_{\text{net}}) = -\frac{1}{2}\varphi^2 + \beta_{\text{net}}\varphi$, monotone on $[-1, 1]$ for $\beta_{\text{net}} \leq 1$. The integral is dominated by the typical maximum of M Gaussian samples $\varphi_{\text{max}}^{(G)}(\alpha) = \sqrt{2\alpha}$; at this edge $g_G(\varphi_{\text{max}}^{(G)}) = \beta_{\text{net}}\sqrt{2\alpha} - \alpha$, so $N^{-1} \ln S \rightarrow \beta_{\text{net}}\sqrt{2\alpha}$ and $u_{\text{noise}}^{(G)} = 1 - \sqrt{2\alpha}$. Solving $u_{\text{noise}}^{(G)} = f_{\text{ret}}(T)$:

$$\alpha_c^G(T) = \frac{1}{2}[1 - f_{\text{ret}}(T)]^2, \quad \alpha_c^G(0) = \frac{1}{2}. \quad (9)$$

β_{net} cancels — both retrieval and spurious-extreme energies are $N \cdot \beta_{\text{net}}$ times an overlap, and the β_{net}^{-1} prefactor in H removes it. This is Ramsauer's threshold.

Exact density, support edge. Keep $f(\varphi)$ but use the support edge of the exact density $\varphi_{\text{max}}(\alpha) = \sqrt{1 - e^{-2\alpha}}$ (from $M \cdot \Pr_f(\varphi > \varphi_{\text{max}}) = 1$). At this edge $\frac{1}{2} \ln(1 - \varphi_{\text{max}}^2) = -\alpha$, so $g(\varphi_{\text{max}}) = -\alpha + \beta_{\text{net}} \varphi_{\text{max}}$, $N^{-1} \ln S \rightarrow \beta_{\text{net}} \varphi_{\text{max}}(\alpha)$ and $u_{\text{noise}}^{(\text{ex})} = 1 - \varphi_{\text{max}}(\alpha)$. Solving $1 - \varphi_{\text{max}}(\alpha_c^{\text{bd}}) = f_{\text{ret}}(T)$:

$$\alpha_c^{\text{bd}}(T) = -\frac{1}{2} \ln[1 - (1 - f_{\text{ret}}(T))^2], \quad \alpha_c^{\text{bd}}(0) = +\infty. \quad (10)$$

This is the formula that works for the LSR kernel where g is monotone on the support [2]; for LSE it is correct only when the interior saddle φ^* lies outside the support $[0, \varphi_{\text{max}}(\alpha)]$, i.e. for $\alpha < -\frac{1}{2} \ln(1 - \varphi^{*2}) \approx 0.241$ at $\beta_{\text{net}} = 1$. Above this threshold the interior saddle (8) takes over. Both (9) and (10) are β_{net} -independent for the same reason: the support edge replaces the β_{net} -dependent saddle.

Cusp picture. At fixed φ_1 the spurious sum evaluates to $e^{N(\alpha + g_{\max}(\beta_{\text{net}}))}$ to leading order, independent of φ_1 . Inside the LSE log-sum, $\beta_{\text{net}}^{-1} \ln[e^{\beta_{\text{net}} N \varphi_1} + S] \rightarrow \max(\varphi_1, [\alpha + g_{\max}(\beta_{\text{net}})]/\beta_{\text{net}})$ at large N , so the leading-order free energy per spin reads

$$\frac{F(\varphi_1)}{N} = -\max\left(\varphi_1, \frac{\alpha + g_{\max}(\beta_{\text{net}})}{\beta_{\text{net}}}\right) - \frac{T}{2} \ln(1 - \varphi_1^2), \quad (11)$$

with a kink (“cusp”) at $\varphi_c(\alpha, \beta_{\text{net}}) = [\alpha + g_{\max}(\beta_{\text{net}})]/\beta_{\text{net}}$. As T rises the basin minimum $\phi_{\text{eq}}(T)$ slides toward the cusp; at $\phi_{\text{eq}}(T) = \varphi_c$ the local minimum disappears, defining the single-pattern spinodal

$$\alpha_c^{\text{sp}}(\beta_{\text{net}}, T) = \beta_{\text{net}} \phi_{\text{eq}}(T) - g_{\max}(\beta_{\text{net}}), \quad T_{\text{sp}}(\alpha, \beta_{\text{net}}) = \frac{\beta_{\text{net}}^2 - (\alpha + g_{\max}(\beta_{\text{net}}))^2}{\beta_{\text{net}} (\alpha + g_{\max}(\beta_{\text{net}}))}. \quad (12)$$

$\alpha_c^{\text{sd}}(\beta_{\text{net}}, T)$ and $\alpha_c^{\text{sp}}(\beta_{\text{net}}, T)$ meet at the endpoint $(\beta_{\text{net}} - g_{\max}(\beta_{\text{net}}), 0)$.

Examples and the $\beta_{\text{net}} \rightarrow \infty$ limit. Four cases of (6), (7), and (8) at $T = 0$:

β_{net}	$\varphi^*(\beta_{\text{net}})$	$g_{\max}(\beta_{\text{net}})$	$\alpha_c(\beta_{\text{net}}, 0)$	note
0.5	$\sqrt{2} - 1 \approx 0.414$	≈ 0.113	≈ 0.387	—
1	$(\sqrt{5} - 1)/2 \approx 0.618$ (golden ratio)	≈ 0.3774	≈ 0.6226	MC value
2	$(\sqrt{17} - 1)/4 \approx 0.781$	≈ 1.091	≈ 0.909	—
$\rightarrow \infty$	$\rightarrow 1 - 1/(2\beta_{\text{net}})$	$\beta_{\text{net}} - \frac{1}{2}(\ln \beta_{\text{net}} + 1)$	$\frac{1}{2}(\ln \beta_{\text{net}} + 1)$	log growth

Capacity grows logarithmically with β_{net} . The Ramsauer prediction $\alpha_c^{\text{G}} = 1/2$ is β_{net} -independent: its derivation replaces the integrand in (4) by its support edge $\sqrt{2\alpha}$, and the factor β_{net} then cancels between retrieval and spurious terms. The interior saddle restores the β_{net} dependence; the β_{net} -independence of the Gaussian-edge answer is itself the signature of the support-edge replacement. The MC numerics in Sections 3 and 4 fix $\beta_{\text{net}} = 1$, where $\alpha_c(1, 0) \approx 0.6226$ is the value we verify. Figure 1 illustrates (11) at $\alpha = 0.50$, $\beta_{\text{net}} = 1$.

3 Monte Carlo schemes

All chains run on the $\sqrt{N}$ -sphere at $\beta_{\text{net}} = 1$ with Metropolis updates. The variants differ in how they handle the $M = e^{\alpha N}$ patterns.

Direct. Store all M patterns explicitly and evaluate (1) over all of them at every step. Memory and compute cost $\sim MN$. Feasible up to $N \approx 25$ at $\alpha \approx 0.7$ ($M \approx 4 \times 10^7$); impossible at $N = 100$, $\alpha = 0.6$ ($M \approx 10^{26}$).

Truncated. Generate only the patterns with overlap $\varphi_{1\mu} > \varphi_{\text{keep}}$ to the target ξ^1 . The cutoff is chosen so the expected count above it equals a budget $K \sim 10^4$; the actual count is Poisson with mean $M \cdot \Pr(\varphi > \varphi_{\text{keep}})$, and the K overlap values are drawn from the Beta tail. The patterns below φ_{keep} are not generated; their contribution enters the LSE log-sum as the analytic constant $\ln C_{\text{bulk}} = \alpha N + \ln \int_{-1}^{\varphi_{\text{keep}}} f(\varphi) e^{N\varphi} d\varphi$. The retained set is fixed once per disorder. Truncated MC at $N = 25$ matches direct MC on the basin boundary across the full α range we measure, so the $\varphi < \varphi_{\text{keep}}$ patterns are genuinely unnecessary for the basin physics, not silently dropped. We use truncated for the wide heatmap.

Hybrid. Same kept-tail + analytic-bulk decomposition as truncated, with two changes: the cutoff is fixed *at the cusp position* $\varphi_c(\alpha, \beta_{\text{net}}) = [\alpha + g_{\max}(\beta_{\text{net}})]/\beta_{\text{net}}$ from (11) rather than chosen by budget, and the bulk integral is evaluated by its saddle (7) instead of numerically. Patterns with $\varphi_{1\mu} > \varphi_c$ are kept live; the contribution of those with $\varphi_{1\mu} < \varphi_c$ enters the LSE

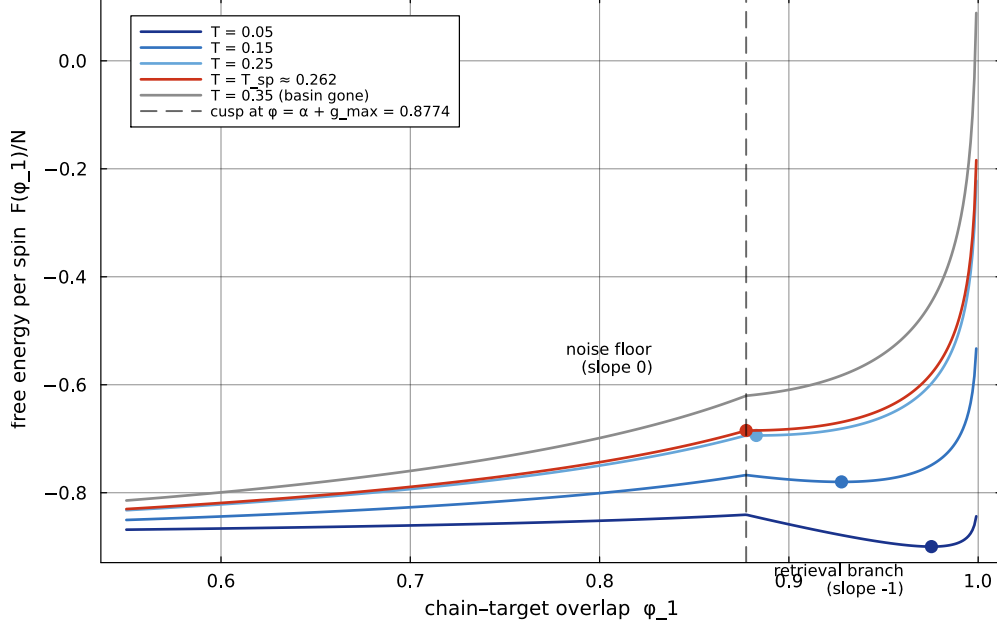


Figure 1: Leading-order free energy per spin $F(\varphi_1)/N$ from (11) at $\alpha = 0.50$, $\beta_{\text{net}} = 1$, for several T . The cusp at $\varphi_c = \alpha + g_{\text{max}} \approx 0.877$ is the boundary between the retrieval branch (slope -1) and the noise-floor branch (slope 0). Dots: basin minima $\phi_{\text{eq}}(T)$. At $T = T_{\text{sp}}(0.50, 1) \approx 0.262$ the minimum reaches the cusp; above this T the basin no longer exists.

log-sum as the $\mathbf{x}$ -independent number $M e^{Ng_{\text{max}}(\beta_{\text{net}})}$ (state-independent by rotational symmetry of random patterns, after averaging over their orthogonal directions to $\boldsymbol{\xi}^1$).

The live count K_{cusp} is drawn from a Poisson with mean $M \cdot \Pr_f(\varphi > \varphi_c)$. Live overlap values are sampled from the spherical density (2) restricted to $\varphi > \varphi_c$. In the rescaled variable $y = (1 + \varphi)/2$ on $[0, 1]$, (2) becomes $\text{Beta}((N-1)/2, (N-1)/2)$ — the change of variable $(1 - \varphi^2)^{(N-3)/2} d\varphi \propto [y(1-y)]^{(N-3)/2} dy$ — so we sample

$$y \sim \text{Beta}\left(\frac{N-1}{2}, \frac{N-1}{2}\right) \text{ truncated to } y \in [(1 + \varphi_c)/2, 1], \quad \varphi_{1\mu} = 2y - 1, \quad (13)$$

which standard libraries provide directly. Each live pattern is constructed from $\varphi_{1\mu}$ and a uniform direction u_μ on the $(N-2)$ -sphere perpendicular to $\boldsymbol{\xi}^1$: $\boldsymbol{\xi}^\mu = \sqrt{N} (\varphi_{1\mu}, \sqrt{1 - \varphi_{1\mu}^2} u_\mu)$ in the basis where $\boldsymbol{\xi}^1 = (\sqrt{N}, 0, \dots, 0)$. The live set is fixed once per disorder.

Metropolis runs on the full spin $\mathbf{x} \in \sqrt{N} \cdot S^{N-1}$: Gaussian proposal $\mathbf{x}' = \mathbf{x} + \sigma \boldsymbol{\eta}$ projected back to $|\mathbf{x}'| = \sqrt{N}$; energy $H(\mathbf{x}) = -\beta_{\text{net}}^{-1} \ln[e^{\beta_{\text{net}} \boldsymbol{\xi}^1 \mathbf{x}} + \sum_{\mu=2}^{K_{\text{cusp}}+1} e^{\beta_{\text{net}} \boldsymbol{\xi}^\mu \mathbf{x}} + M e^{Ng_{\text{max}}(\beta_{\text{net}})}]$. Cost per step $O(K_{\text{cusp}} \cdot N)$.

Crucially the bulk being a constant in $\mathbf{x}$ does not make the free energy constant in φ_1 : the chain still feels the sphere geometry, and below the cusp (11) gives $F(\varphi_1)/N = -\varphi_c - (T/2) \ln(1 - \varphi_1^2)$, which is minimised at $\varphi_1 = 0$. Once the chain crosses the cusp it slides toward $\varphi_1 = 0$ under this entropic attraction; that slide is what we use to detect basin escape (Sec. 4).

At $N = 100$ the spherical density falls fast enough that the expected K_{cusp} at the cusp is below 1 for every $\alpha \in [0.5, 0.62]$ we measure. The algorithm therefore keeps only $\boldsymbol{\xi}^1$ explicit and the bulk constant carries the rest — the chain samples the leading-order partition function (11) directly, with no $1/N$ subleading corrections.

Why naive “smart” fails. A faster variant — “smart” — refreshes its K kept patterns every N_{REFRESH} steps from a cone around the current chain. This breaks the quenched-disorder assumption. As the chain wanders, fresh patterns are minted at every refresh; the total set of distinct patterns seen along one trajectory exceeds M , and the chain effectively faces a larger

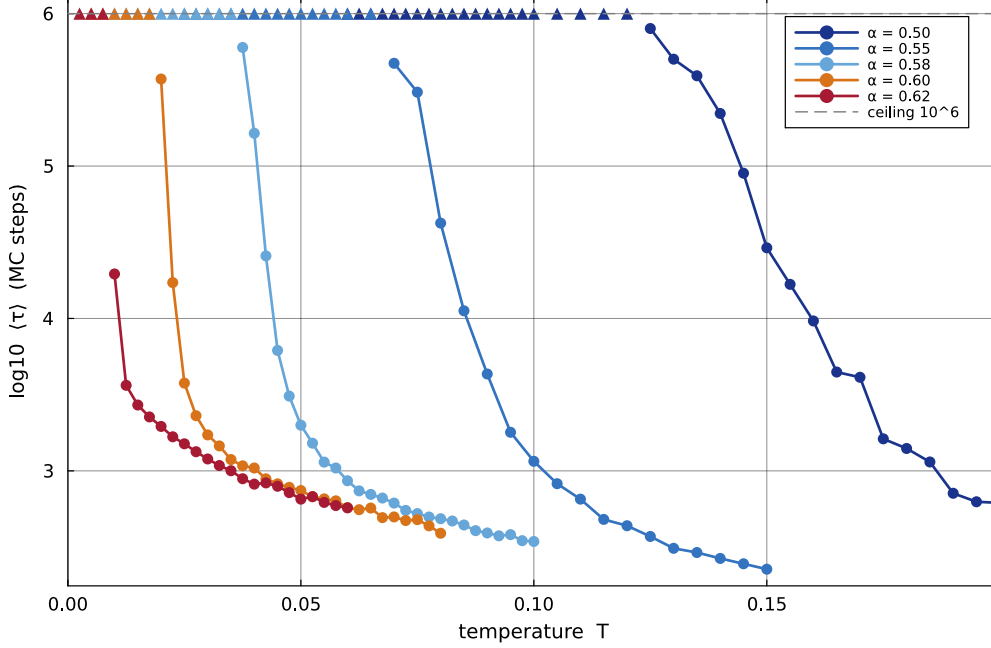


Figure 2: Mean basin escape time $\langle \tau \rangle$ vs T at $N = 100$ from N-dim hybrid MC, 32 disorders, $\beta_{\text{net}} = 1$. Five α values: 0.50, 0.55, 0.58, 0.60, 0.62. Filled circles: mean over disorders that escaped within 10^6 MC steps. Upward triangles at $\log_{10} \tau = 6$: cells where no disorder escaped (lower bound only). The escape curve slides toward $T = 0$ as $\alpha \rightarrow 1 - g_{\text{max}} \approx 0.6226$.

pattern set than the model defines. Near the retrieval basin the extra competitors destroy the basin. Both truncated and hybrid avoid the problem: the retained set is fixed once per disorder, no refresh.

4 Results

We measure two observables at $N = 100$: the mean first-passage escape time τ from the retrieval basin, and the steady-state $\langle \varphi_1 \rangle - \phi_{\text{eq}}(T)$ residual.

Escape time. Figure 2 reports $\log_{10} \langle \tau \rangle$ vs T at $N = 100$ for five α values from the N-dim hybrid MC, averaged over 32 disorders, with a budget cap at 10^6 MC steps. Each curve is a basin escape profile: at low T the basin is stable for the full budget (upward triangles at the ceiling); above a critical $T_{\text{esc}}(\alpha)$ the escape time drops by several orders of magnitude over a T window of width ~ 0.01 . T_{esc} moves to lower T as α rises; extrapolating to $\tau \rightarrow \infty$ at $T = 0$ gives a critical load near $\alpha = 1 - g_{\text{max}} \approx 0.6226$.

Residual heatmap. Figure 3 shows $\langle \varphi_1 \rangle - \phi_{\text{eq}}(T)$ over the (α, T) plane. Background: truncated MC at varying N over $\alpha \in [0.20, 0.70]$. Grey rectangle: N-dim hybrid MC at $N = 100$ on a fine T grid in $\alpha \in [0.50, 0.62]$, $T \in [0, 0.10]$. The basin (white) is bounded by a sharp transition (dark red) that follows the saddle line $\alpha_c^{\text{sd}}(T)$ and pinches at $\alpha \approx 0.6226$, $T = 0$.

5 Discussion

The Ramsauer threshold $\alpha_c = 1/2$ is a Gaussian artefact. The Gaussian density misses the interior saddle of the partition-sum integrand (4) that the exact spherical density places at $\varphi^*(\beta_{\text{net}})$. Including it gives $\alpha_c(\beta_{\text{net}}, 0) = \beta_{\text{net}} - g_{\text{max}}(\beta_{\text{net}})$; at the value $\beta_{\text{net}} = 1$ used in our MC, $\alpha_c \approx$

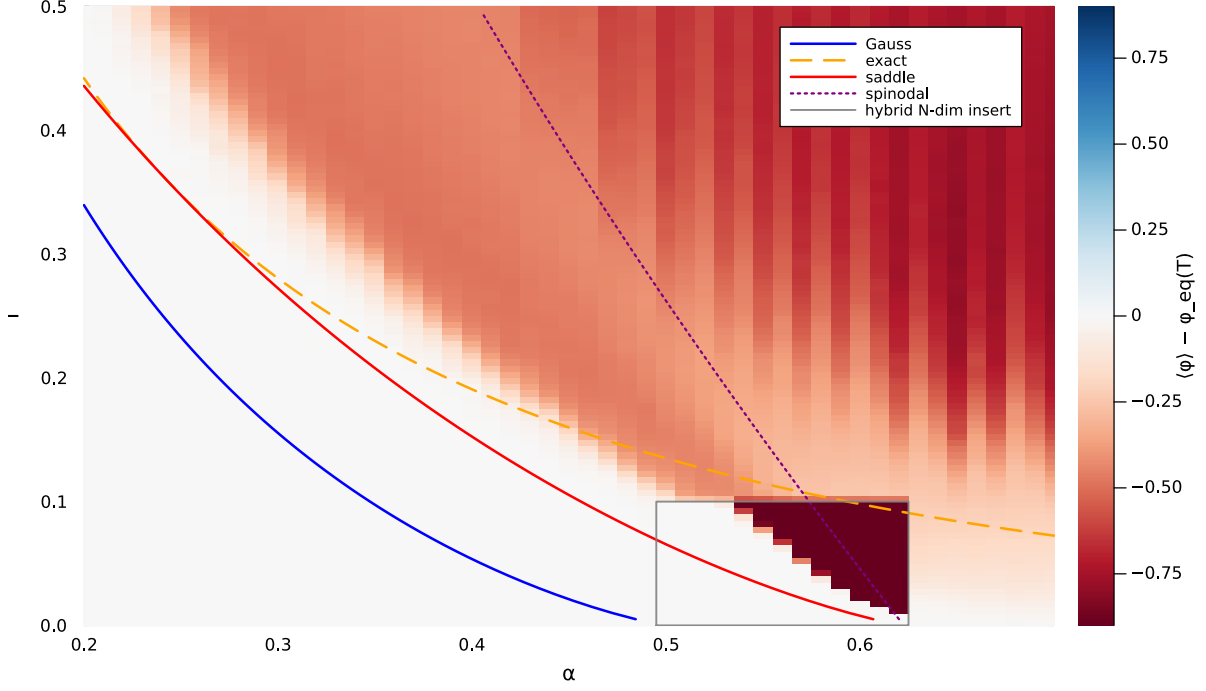


Figure 3: $\langle \varphi_1 \rangle - \phi_{\text{eq}}(T)$ from Monte Carlo over the (α, T) plane. Curves: $\alpha_c^G(T)$ (blue solid, Eq. 9, Gaussian density + support edge); $\alpha_c^{\text{bd}}(T)$ (orange dashed, Eq. 10, exact density + support edge); $\alpha_c^{\text{sd}}(T)$ (red solid, Eq. 8, exact density + interior saddle, this work); $\alpha_c^{\text{sp}}(T)$ (purple dotted, Eq. 12, single-pattern spinodal). Background: truncated MC at varying N over the full α range, linearly interpolated to the fine T grid. Grey rectangle: N-dim hybrid MC at $N = 100$ on the fine T grid. The basin pinches at $\alpha \approx 0.6226$ — Ramsauer’s $\alpha_c^G(0) = 1/2$ undershoots by 0.12 in α , a factor 1.25 in capacity.

0.6226, a factor 1.25 larger than Ramsauer’s $1/2$. The shift is per spin, not asymptotic: it appears at leading order in N and is visible already at $N = 100$. A second signature of the artefact is that Ramsauer’s $1/2$ is β_{net} -independent, while the exact-density (8) boundary is not.

Open items. (i) Hot init. All present runs start the chain at ξ^1 (cold). The basin boundary they trace is the spinodal, not the thermodynamic crossover; hot init at the same grid would delimit the metastability band. (ii) Cusp rounding at finite N . The leading-order $\max(\cdot, \cdot)$ in (11) is rounded on a scale $1/N$ in φ_1 . Near the endpoint $(0.6226, 0)$ the basin offset $\phi_{\text{eq}}(T) - (\alpha + g_{\text{max}})$ falls inside the rounding window, and the leading-order spinodal slope needs an NLO correction. (iii) Heatmap consolidation. Figure 3 stitches three data sources. A clean version would use a single MC variant — truncated with a hybrid insert near the cusp — across the entire (α, T) plane.

References

- [1] Dmitry Krotov and John J. Hopfield. Dense associative memory for pattern recognition. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 29, pages 1172–1180, 2016.
- [2] Tatiana Petrova, Evgeny Polyachenko, and Radu State. Retrieval metastability via Kramers escape and centroid hopping. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. placeholder key; replace at submission.
- [3] Hubert Ramsauer, Bernhard Schäffl, Johannes Lehner, Philipp Seidl, Michael Widrich, Lukas Gruber, Markus Holzleitner, Thomas Adler, David Kreil, Michael K. Kopp, Günter Klam-

bauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021.